Problem Set 10

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Overall the algorithms performed pretty similarly to each other, with test accuracies ranging from 0.843 to 0.868. The decision tree came out on top and kNN was the worst, but honestly the gap between all of them is pretty small so it is hard to say any one algorithm is clearly better than the others for this dataset.

1 Model Accuracies

Table 1: Optimal Tuning Parameters and Out-of-Sample Accuracy

Algorithm	CV Acc.	Test Acc.	λ	Cost Complexity	Tree Depth	Min n	Hidden Units	Neighbors	Cost	σ
Logit	0.846	0.853	1×10^{-10}	–	–	–	–	–	–	–
Tree	0.859	0.868	–	0.001	15	10	–	–	–	–
NNet	0.846	0.852	1.00	–	–	–	5	–	–	–
kNN	0.838	0.843	–	–	–	–	–	30	–	–
SVM	0.852	0.864	–	–	–	–	–	–	1.00	0.25

Note: Accuracy reported on held-out test set (20% of sample). Models tuned via 3-fold cross-validation on training set.